Chronology

a chronology
of events, insights, and discoveries

1687	Newton publishes his *Principia*, in which are formulated his concepts of absolute space and time, and his laws of motion and laws of gravity. [Ch. 1]
1783&1795	Michell and Laplace, using Newton's laws of motion, gravity, and light, formulate the concept of a Newtonian black hole. [Ch. 3]
1864	Maxwell formulates his unified laws of electromagnetism. [Ch. 1]
1887	Michelson and Morley show, experimentally, that the speed of light is independent of the velocity of the Earth through absolute space. [Ch. 1]
1905	Einstein shows that space and time are relative rather than absolute, and formulates the special relativistic laws of physics. [Ch. 1]
	Einstein shows that electromagnetic waves behave under some circumstances like particles, thereby initiating the concept of wave/particle duality that underlies quantum mechanics. [Ch. 4]
1907	Einstein, taking his first steps toward general relativity, formulates the concept of a local inertial frame and the equivalence principle, and deduces the gravitational dilation of time. [Ch. 2]